Q1. Using the linear combinations of atomic orbitals (LCAO), obtain wavefunctions for the bonding and anibonding molecular orbitals (MOs) of H_2^+ molecule.

A1

H_2^+ : No. of electrons = 1

Let's consider $H - H$

A-atom B-atom

ψ_{1s}^A ψ_{1s}^B

Apply LCAO

$$\psi_{Bonding} = C_A \psi_{1s}^A + C_B \psi_{1s}^B$$

$$\psi_{Anti\ bonding} = C_A \psi_{1s}^A - C_B \psi_{1s}^B$$

Q2. A homonuclear diatomic system has the ground-state molecular orbital (MO) configuration $1\sigma_g^2 1\sigma_u^2 2\sigma_g^2 2\sigma_u^2 3\sigma_g^2 1\pi_u^4 1\pi_g^2$. Answer the following:

(a) What is the number of bonding electrons?

(b) Find the bond order.

(c) What would you expect the effect to be on the bond dissociation energy of this molecule when an electron is removed (a) from the $1\pi_g$ MO ? (b) from the $3\sigma_g$ MO?

A2

(a) No. of bonding electrons = 10

No. of antibonding electrons = 6

(b) $B.O. = \frac{10-6}{2} = \frac{4}{2} = 2$

(c) (a) Ans, $1\pi_g$ is antibonding, $BO = \frac{10-5}{2} = \frac{5}{2} = 2.5$

After removing one electron as bond order has increased, bond dissociation energy will increase

(b) $3\sigma_g$ is bonding, $BO = \frac{9-6}{2} = \frac{3}{2} = 1.5$

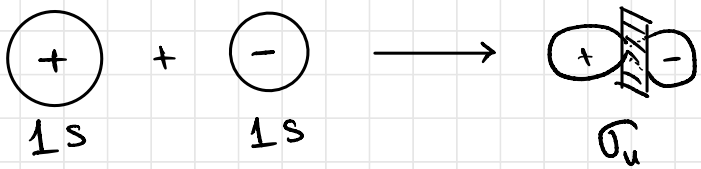
After removing one electron as bond order has decreased bond dissociation energy will decrease

Q3. Indicate whether you expect each of the following homonuclear diatomic MOs to be bonding or antibonding. Sketch the MO in each case:

(a) σ_u (b) π_u (c) π_g

A3

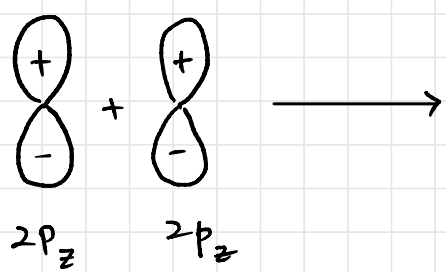
a) σ_u Antibonding

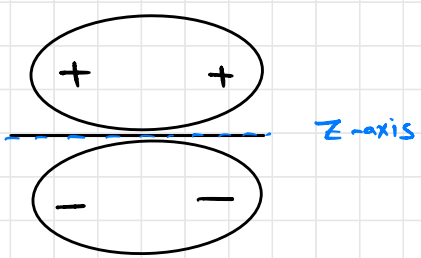


σ_u

b) π_u Bonding

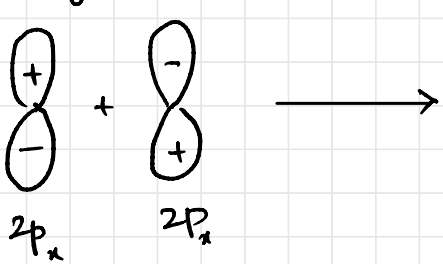
Considering "z" as internuclear axis

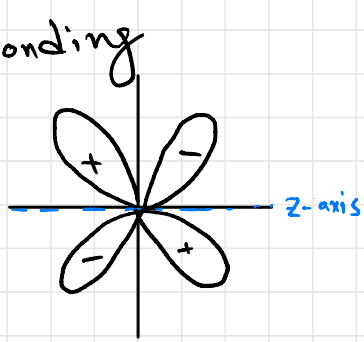




π_u

c) π_g Antibonding





Q4. Arrange bond dissociation energies of O_2, N_2, O_2^+, and N_2^+ in increasing bond order.

A4

Bond Order $=$

$$\frac{\text{no. of bonding electrons} - \text{no. of antibonding electrons}}{2}$$

For N_2 : No. of electrons $= 7 \times 2 = 14$

Electronic Configuration

$$1\sigma_g^2\, 1\sigma_u^2\, 2\sigma_g^2\, 2\sigma_u^2\, 1\pi_u^{x\,2}\, 1\pi_u^{y\,2}\, 3\sigma_g^2\, 1\pi_g^x\, 1\pi_g^y\, 3\sigma_u$$

$$\Rightarrow \text{B.O.} = \frac{10-4}{2} = 3$$

For O_2 : No. of electrons $= 8 \times 2 = 16$

Electronic Configuration

$$1\sigma_g^2\, 1\sigma_u^2\, 2\sigma_g^2\, 2\sigma_u^2\, 3\sigma_g^2\, 1\pi_u^{x\,2}\, 1\pi_u^{y\,2}\, 1\pi_g^{x\,1}\, 1\pi_g^{y\,1}\, 3\sigma_u$$

$$\Rightarrow \text{B.O.} = \frac{10-6}{2} = 2$$

For N_2^+ : No. of electrons = $7 \times 2 = 14$

Electronic Configuration

$1\sigma_g^2\ 1\sigma_u^2\ 2\sigma_g^2\ 2\sigma_u^2\ 1\pi_u^{x\,2}\ 1\pi_u^{y\,2}\ 3\sigma_g^1\ 1\pi_g^x\ 1\pi_g^y\ 3\sigma_u$

$\Rightarrow$ B.O. $= \frac{9-4}{2} = 2.5$

For O_2^+ : No. of electrons = $8 \times 2 = 16$

Electronic Configuration

$1\sigma_g^2\ 1\sigma_u^2\ 2\sigma_g^2\ 2\sigma_u^2\ 3\sigma_g^2\ 1\pi_u^{x\,2}\ 1\pi_u^{y\,2}\ 1\pi_g^{x\,1}\ 1\pi_g^y\ 3\sigma_u$

$\Rightarrow$ B.O. $= \frac{10-5}{2} = 2.5$

Increasing bond order :

$$O_2 < O_2^+ = N_2^+ < N_2$$

Increasing bond dissociation energies :

$$O_2 < O_2^+ = N_2^+ < N_2$$

Q5. A homonuclear diatomic MO of π_u symmetry is to be expressed as a linear combination of AOs centered on the nuclei laying on the z-axis. Which of the AOs in the following list can contribute to the MO?

(a) 1s (b) 2s (c) $2p_x$ (d) $2p_y$ (e) $2p_z$ (f) $3p_x$ (g) $3p_y$ (h) $3p_z$

A5

(c) $2p_x$

(d) $2p_y$

(f) $3p_x$

(g) $3p_y$